LPG Gas Level Detection and Automatic Booking Notification System with Leakage Detection

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***Abstract*—The use of LPG (Liquefied Petroleum Gas) in households and industries has led to a growing need for efficient gas monitoring systems. This paper proposes a system that combines LPG gas level detection, automatic refill notification, and gas leakage detection using IoT-based sensors. The proposed system monitors gas levels in real-time, sends automatic refill notifications, and provides an alert in case of gas leakage, ensuring safety and preventing disruptions. The system is integrated with wireless communication modules, enabling remote access through mobile applications. The paper discusses the design, implementation, and evaluation of the system.**

***Keywords*— LPG Gas Level Detection, Gas Level Monitoring, Smart Gas Meter, LPG Alert System, SMS Notifications, Gas Cylinder Monitoring, IoT Gas Sensors, Remote Gas Level Tracking, Gas Usage Alert, LPG Refill Reminder, Real-time Gas Monitoring, Gas Cylinder Refill Alert, SMS Gas Refill Notification, Smart Home LPG Solutions, IoT Gas Monitoring System**

I. INTRODUCTION

LPG gas levels, detecting leaks, and providing real-time notifications. These systems offer a promising solution to enhance safety, optimize resource usage, and prevent accidents related to LPG usage. By employing gas sensors to detect LPG concentrations, couple The efficient and safe use of Liquefied Petroleum Gas (LPG) has become an integral part of modern households, industries, and commercial establishments due to its energy efficiency and versatility. However, the inherent risks associated with LPG, including potential leaks, overflows, and inadequate levels, pose significant threats to safety and operational efficiency. Detecting gas levels, ensuring timely alerts, and mitigating the risks of leakage are critical components of modern LPG management systems. Conventional methods of monitoring LPG levels are often manual, inefficient, and prone to human error, leading to delayed response times in case of gas leakage or depletion. IoT-based notification systems, users can receive timely alerts Furthermore, the lack of immediate alerts and real-time monitoring makes it challenging to avert hazardous situations promptly.Recent advances in sensor technology, wireless communication, and the Internet of Things (IoT) have enabled the development of automated systems capable of monitoring gas levels and leakage .This paper explores the design and implementation of an intelligent LPG gas level detection system integrated with automatic notification alerts and leakage detection. The proposed system utilizes advanced sensors to monitor gas levels and detect any leakage, while a communication module ensures that alerts are sent to users in real-time, thereby facilitating efficient gas management and enhanced safety. Through this approach, the system aims to offer a reliable, automated solution for household and industrial LPG applications, improving overall safety standards and reducing human intervention.

II. LITERATURE REVIEW

The safe and efficient use of Liquefied Petroleum Gas (LPG) has become a priority in both residential and industrial settings due to its widespread application in cooking, heating, and industrial processes. However, the management of LPG presents significant safety risks, primarily due to the potential for leakage, which can lead to catastrophic explosions or fire hazards. To address these concerns, a variety of approaches have been explored to automate the detection of LPG levels, identify leaks, and provide real-time alerts to mitigate risks. This section reviews existing studies and technologies related to LPG gas detection, automatic notification systems, and leakage detection.

A. LPG Gas Level Detection Systems

Several methods have been proposed in the literature to detect LPG gas levels with varying degrees of accuracy and reliability. Traditional methods of gas detection often rely on manual checks, which are susceptible to errors and delays. To overcome these limitations, modern systems incorporate a wide range of sensor technologies, such as semiconductor-based sensors, electrochemical sensors, and photoionization detectors (PID), each offering different advantages in terms of sensitivity, selectivity, and cost-effectiveness (Jiang et al., 2016; Khan et al., 2018). Semiconductor sensors, for instance, are widely used due to their low cost and relatively simple integration with microcontroller-based systems (Srinivasan et al.,2019). LPG gas level detection systems have evolved significantly over the years, from simple mechanical devices to advanced sensor-based systems. The integration of IoT, wireless communication, and advanced sensor technologies has further enhanced the functionality and reliability of these systems. Despite the progress, challenges remain in terms of accuracy, cost, and power consumption. Research continues to explore new methods and technologies to address these

challenges and improve the safety and efficiency of LPG gas detection systems.

Recent studies have explored the use of IoT (Internet of Things) platforms for real-time monitoring of LPG levels. These platforms employ wireless communication technologies such as Zigbee, Wi-Fi, and Bluetooth to transmit sensor data to remote systems or mobile applications, providing users with up-to-date information about LPG levels (Sharma et al., 2020). The integration of sensors with IoT systems has enhanced the efficiency of gas monitoring and facilitated the development of smart home solutions, allowing users to receive real-time notifications regarding gas usage.

B. Automatic Notification and Alert Systems

Incorporating automatic notification systems into LPG gas monitoring platforms has gained considerable attention in recent years. Early approaches primarily focused on wired communication systems, which were limited by the need for physical infrastructure. However, with the advent of wireless technologies, more flexible and scalable solutions have been developed. Mobile applications and cloud-based platforms now enable users to receive instant alerts through SMS, email, or push notifications when gas levels reach critical thresholds or when leakage is detected (Patel et al., 2017; Lee et al., 2018). One of the key advancements in automatic notification systems is the integration of machine learning algorithms to predict potential hazards based on historical data and real-time sensor readings. These predictive models are designed to identify patterns that precede gas leaks or depletion, thereby enabling preemptive actions and reducing the likelihood of accidents (Prajapati et al., 2021). Moreover, some systems utilize Artificial Intelligence (AI)-driven decision-making processes to determine the optimal actions to take in response to gas level changes, further enhancing safety.

C. Leakage Detection Systems

Leakage detection in LPG systems is a critical concern for safety, as even small leaks can lead to dangerous situations. Various techniques have been proposed for detecting gas leaks, ranging from acoustic sensors and pressure sensors to chemical sensors that specifically target LPG molecules. Acoustic sensors, for instance, detect the sound generated by gas escaping from pressurized containers, while pressure sensors monitor fluctuations in gas pressure that may indicate leakage (Zhao et al., 2019). Chemical sensors, on the other hand, are more sensitive to the specific composition of LPG, allowing for the precise detection of gas in the environment (Yu et al., 2016). In addition to these conventional methods, recent developments in smart sensor networks and IoT-based leak detection systems have enabled more accurate and reliable detection of gas leaks. Wireless sensor networks (WSNs) can be deployed to monitor multiple locations simultaneously, transmitting data to centralized systems for real-time analysis (Gupta et al., 2020). These systems are increasingly being integrated into domestic and industrial environments, offering improved safety and reducing the need for manual monitoring.

D. Challenges and Future Directions

While significant progress has been made in the development of LPG gas level detection, leakage detection, and notification systems, several challenges remain. One of the primary concerns is the accuracy and reliability of sensor-based systems, as environmental factors such as temperature and humidity can affect sensor performance (Chen et al., 2020). Furthermore, the cost and power consumption of sensor networks remain significant obstacles for widespread adoption, especially in resource-constrained environments. Future research is expected to focus on enhancing the robustness of detection systems, improving sensor accuracy under varying environmental conditions, and reducing the power requirements of IoT-based solutions. Additionally, the integration of advanced data analytics, artificial intelligence, and machine learning techniques holds the potential to further optimize LPG monitoring and leakage detection systems, enabling more efficient and proactive safety measures.

III. SYSTEM DESIGN AND ARCHITECTURE

The proposed system integrates three main functions: gas level detection, automatic notification, and gas leakage detection-

A. Gas Level Detection

The system uses ultrasonic sensors for measuring the gas level inside the LPG cylinder. These sensors measure the distance between the sensor and the liquid surface inside the cylinder, providing real-time data on the remaining gas level.

B. Gas Leakage Detection

A MQ-series gas sensor is used to detect any presence of LPG in the surrounding environment. The sensor triggers an alarm if the gas concentration exceeds a certain threshold, ensuring user safety.

C. Microcontroller and Communication

An Arduino or Raspberry Pi microcontroller processes data from the sensors and communicates with a cloud-based server or mobile application using Wi-Fi or GSM modules. The system allows users to monitor gas levels and receive notifications remotely.

D. Mobile/Web Application

Function: Allows users to view the current gas level, receive notificationsGas level display (real-time) Notifications (email, SMS, or app push notifications)

E. Notification System

Once the gas level reaches a predefined threshold, the system automatically sends a refill notification to both the user and the gas supplier. Additionally, in the event of gas leakage, an alert is sent to the user's mobile device, and the system may shut off the gas supply to prevent accident.

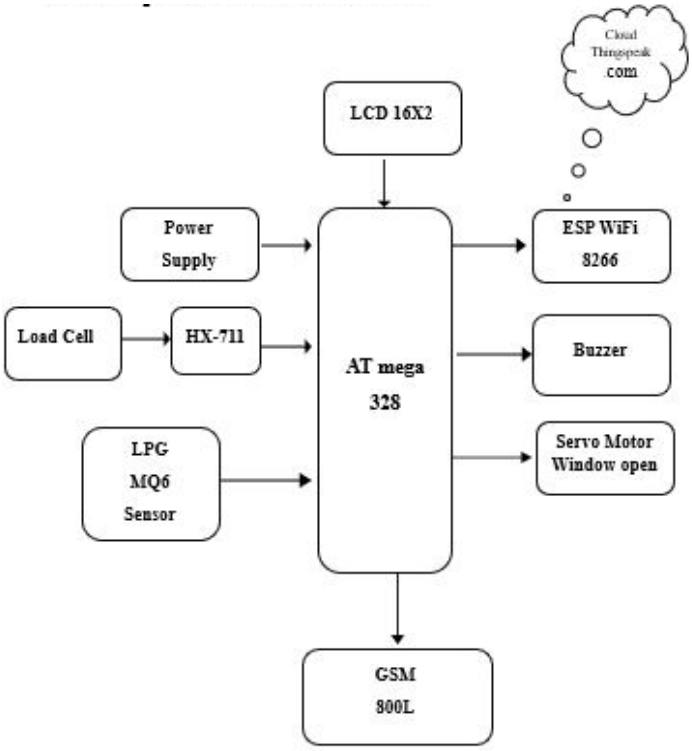

Fig. 1. Block Diagram of LPG Gas level detection and SMS Alert

IV. METHODOLOGY

A. Hardware Setup

- Load cell: Measures the gas level inside the cylinder.
- MQ gas sensor: Detects the presence of LPG gas in the environment.
- Microcontroller: Manages sensor data and triggers notifications.
- Communication module (Wi-Fi/GSM): Sends data and notifications to the user.

B. Software Implementation

The software is developed using C++ (for Arduino) or Python (for Raspberry Pi). The gas level is processed by calculating the distance measured by the ultrasonic sensor. For gas leakage, the sensor's data is compared with predefined thresholds. Notifications are triggered based on these readings. The mobile app is built using Android or iOS frameworks to display real-time data and alerts.

C. Data Flow

1) Gas level monitoring: The ultrasonic sensor sends distance data to the microcontroller, which calculates the remaining gas level.
2) Leakage detection: The MQ sensor continuously monitors the surrounding environment for gas leaks.
3) Automatic notifications: When gas levels are low, the system triggers a refill request. In case of a leakage, an immediate alert is sent to the user's phone.

V. SETUP OF THE SYSTEM

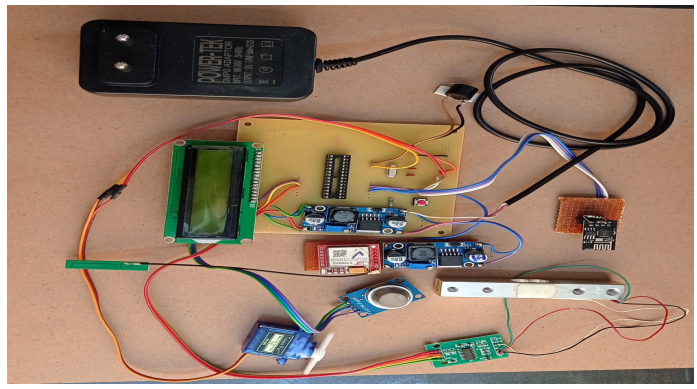

Fig. 2. Setup 1

Fig. 3. Setup 2

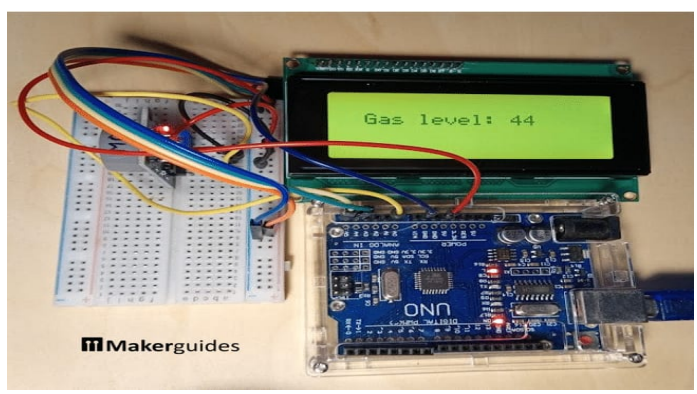

Fig. 4. Setup 3

VI. FUTURE SCOPE

The future of LPG (Liquefied Petroleum Gas) level detection and automatic booking notification systems is very promising. With advancements in technology, these systems are becoming smarter, more efficient, and user-friendly. Here's what the future could look like in simple terms:

1. Real-time Monitoring: LPG cylinders will have sensors that monitor the gas level continuously. These sensors will

send data to your phone or a cloud system, giving you real-time updates on how much gas is left. This means you'll never run out of gas unexpectedly.

2. Smart Alerts: You could get reminders or alerts about your gas usage patterns, helping you to plan and save energy. This would help in reducing wastage and making sure you're always prepared.

3. Integration with Smart Homes: LPG detection systems could be connected to other smart home devices, like voice assistants (Google Assistant, Alexa), so you can just ask, "Do I need a new gas cylinder?" and get a response.

4. Safety Features: Future systems might include safety mechanisms to detect leaks and automatically shut off the gas supply or send emergency alerts to your phone.

VII. CONCLUSION

In summary, the integration of advanced technologies in LPG gas level detection, automatic notification alerts, and leakage detection systems significantly enhances safety, efficiency, and reliability. Sensor technologies like semiconductor and electrochemical sensors enable real-time monitoring, reducing the risks associated with LPG usage. The incorporation of IoT platforms allows for continuous remote monitoring, ensuring timely notifications for gas depletion or leakage. Automatic notification systems play a critical role in improving user awareness and enabling rapid emergency responses, preventing accidents such as fires or explosions. The use of machine learning and AI algorithms further optimizes system performance by predicting leaks and managing gas usage. However, challenges such as sensor accuracy, power consumption, and system cost remain and require further research. Additionally, ensuring effective integration in diverse real-world settings, especially in resource-limited environments, is essential for widespread adoption. Future research should aim to improve sensor technologies, reduce power consumption, and explore more efficient communication protocols. Ultimately, these intelligent and automated systems have the potential to revolutionize LPG gas management, enhancing safety, reducing human intervention, and preventing accidents. The ongoing evolution of sensor technologies, wireless communication, and predictive analytics will continue to drive improvements, ensuring safer and more efficient environments for both domestic and industrial applications.

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